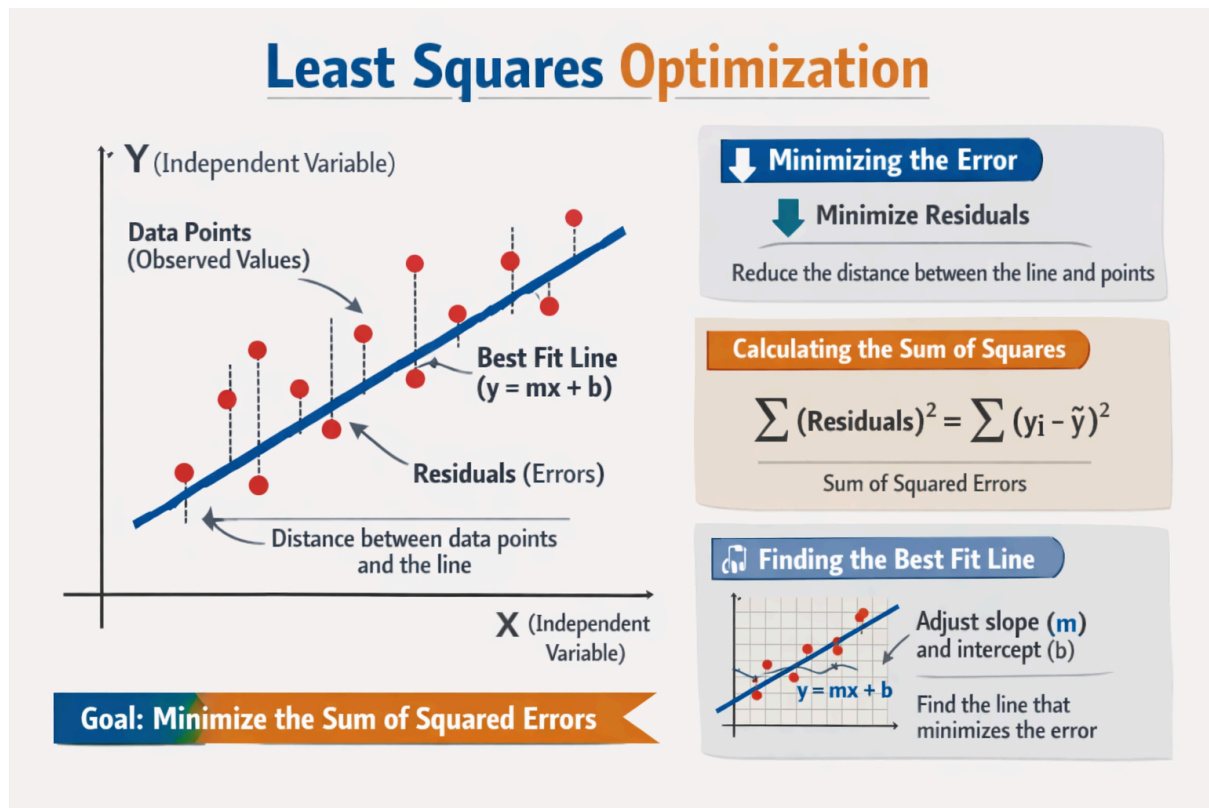


Least Squares Optimization



The key idea is simple:

- Each red point represents observed data.
- The blue line is the regression line we're trying to fit.
- The dashed vertical lines are the residuals (differences between observed and predicted values).
- Least squares optimization adjusts the slope and intercept of the line so that the sum of squared residuals is as small as possible.

This process ensures the line is the “best fit” in terms of minimizing overall error.

Example 1: Linear Regression (Best Fit Line)

Given points: $(1, 1), (-2, -1), (3, 2)$. 

1. Calculate the means of x and y :

$$1. \bar{x} = \frac{1 + (-2) + 3}{3} = \frac{2}{3}$$

$$2. \bar{y} = \frac{1 + (-1) + 2}{3} = \frac{2}{3}$$

2. Calculate the slope (b or m):

$$1. m = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$2. \text{Numerator: } (1 - \frac{2}{3})(1 - \frac{2}{3}) + (-2 - \frac{2}{3})(-1 - \frac{2}{3}) + (3 - \frac{2}{3})(2 - \frac{2}{3}) = \frac{23}{3}$$

$$3. \text{Denominator: } (1 - \frac{2}{3})^2 + (-2 - \frac{2}{3})^2 + (3 - \frac{2}{3})^2 = \frac{38}{3}$$

$$4. m = \frac{23/3}{38/3} = \frac{23}{38}$$

3. Calculate the intercept (a or c):

$$1. c = \bar{y} - m\bar{x} = \frac{2}{3} - \frac{23}{38} \times \frac{2}{3} = \frac{5}{19}$$

4. Equation of the line: $y = \frac{23}{38}x + \frac{5}{19}$. 

Example 2: Data Modeling (Height/Weight)

Find the best-fit line for: Heights $x = [160, 162, 164, 166, 168]$, Weights $y = [52, 55, 57, 60, 61]$

1. **Mean values:** $\bar{x} = 164, \bar{y} = 57$.

2. **Calculations:**

1. $\sum (X - x_i)(Y - y_i) = 46$

2. $\sum (X - x_i)^2 = 40$

3. **Slope (m):** $m = \frac{46}{40} = 1.15$

4. **Intercept (c):** $c = \bar{y} - m\bar{x} = 57 - 1.15 \times 164 = -131.6$

5. **Equation:** $y = 1.15x - 131.6$.

◆ Problem Setup

We want to fit a straight line:

$$y = \beta_0 + \beta_1 x$$

to a given dataset (x_i, y_i) using least squares.

◆ Matrix Formulation

Define:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

The model is:

$$y = X\beta + \epsilon$$

Least squares solution:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

◆ Normal Equations Form

Expanding:

$$\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

Solve this 2×2 system for β_0, β_1 .

◆ Worked Example

Suppose data points: $(1, 2), (2, 3), (3, 5)$.

Step 1: Construct matrices

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad y = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

Step 2: Compute normal equations

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \quad X^T y = \begin{bmatrix} 10 \\ 23 \end{bmatrix}$$

So:

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 10 \\ 23 \end{bmatrix}$$

Step 3: Solve system

Determinant = $3 \cdot 14 - 6 \cdot 6 = 6$.

Inverse:

$$(X^T X)^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$$

Multiply:

$$\hat{\beta} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 10 \\ 23 \end{bmatrix} = \begin{bmatrix} 0.333 \\ 1.5 \end{bmatrix}$$

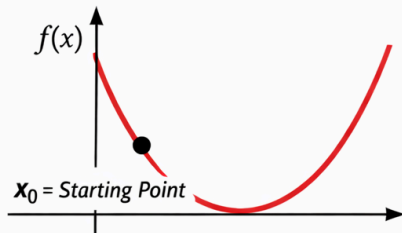
◆ Final Fitted Line

$$y \approx 0.333 + 1.5x$$

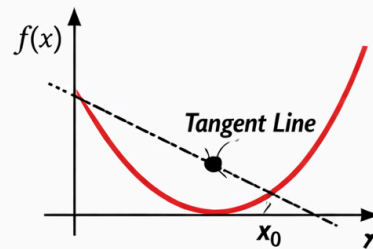
Unconstrained Optimization Technique:

Newton-Raphson Method for Optimization

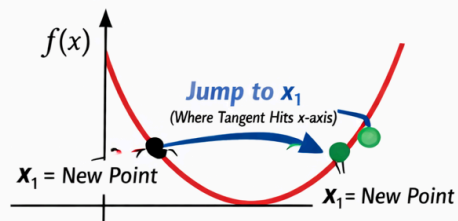
1. Start at Initial Point



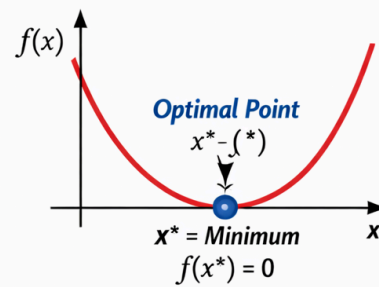
2. Draw Tangent Line



3. Jump to Next Point



4. Reach the Minimum



1. Start at an initial point

- Pick a starting guess x_0 on the curve $f(x)$.
- This is where you begin your search for the minimum.

2. Draw the tangent line

- At x_0 , draw the tangent to the curve.
- The slope of this tangent is $f'(x_0)$.
- The tangent line represents the local linear approximation of the curve.

3. Jump to the next point

- Extend the tangent line until it meets the x-axis.
- The intersection point becomes your next estimate x_1 .
- This step uses the formula:

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

4. Reach the minimum

- Repeat the process until the tangent becomes flat (slope ≈ 0).
- The point where the tangent is horizontal is the **optimal point** x^* .
- At this point, $f'(x^*) = 0$ and $f''(x^*) > 0$, confirming a minimum.

[Refer Link](#) : Pg. 2.22 onwards for numericals